

Electrical Safety & Lockout-Tagout Manual

Document SAF-LOTO-01 · Revision 1.0 · Sample corpus for GraphRAG

1. Purpose and scope

This manual defines the lockout/tagout (LOTO) procedure REQUIRED before any maintenance or inspection task on industrial pumps and their driving motors. It applies to all certified technicians and PRECEDES any task documented in the Industrial Pump Troubleshooting Manual.

2. Hazards addressed

Unexpected energization of a motor is CAUSED_BY residual capacitance, gravity, stored fluid pressure, or improper isolation. These hazards are RELATED_TO the majority of recordable injuries in pump maintenance and can be a SYMPTOM_OF skipping the LOTO sequence below.

3. Required personal protective equipment

PPE for LOTO REQUIRES insulated gloves rated for the system voltage, arc-flash face shield, dielectric footwear, and a personal padlock with a unique key. PPE is PART_OF the daily pre-task checklist and must be inspected before use.

4. Lockout/Tagout procedure

Step 1 — Notify all affected personnel. Step 2 — Identify all energy sources (electrical, hydraulic, pneumatic). Step 3 — Shut down the equipment using normal stopping procedure. Step 4 — Isolate each energy source at its disconnect. Step 5 — Apply your personal lock and tag at every isolation point. Step 6 — Release stored energy (bleed pressure, discharge capacitors). Step 7 — Verify zero energy with a calibrated meter before starting work. Skipping any step is CAUSED_BY time pressure and is the top contributor to near-miss incidents.

5. Restoring service

After maintenance, restoration PRECEDES the next production run and REQUIRES the following: clear the work area, reinstall guards, remove personal locks (only the owner may remove their own lock), notify operators, and re-energize the system at the disconnect. Verify normal operation before leaving the area.

6. Incident response

If an unexpected release of energy occurs, evacuate the zone, isolate the source from a safe distance, and notify the shift supervisor. Document the event within 24 hours; root-cause analysis is PART_OF the continuous improvement program and is RELATED_TO updates of this manual.

7. Emergency contacts

Role	Name	Phone	Availability
Shift Supervisor	On duty	Ext. 101	24x7
EHS Officer	P. Ramanathan	Ext. 215	Mon-Fri 08:00-18:00
Plant Electrician	On call	Ext. 330	24x7
External Emergency	Local fire dept	112	24x7